

VEER SURENDRA SAI UNIVERSITY OF TECHNOLOGY

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Subject: Python

Branch: Mechanical Engineering

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PROBLEM STATEMENT:

BEAM DEFLECTION ANALYSIS USING PYTHON (MATPLOTLIB & SCIPY)

Structural analysis is essential for predicting how beams deform under load. Manual calculation of beam deflection using classical equations is time-consuming and prone to error. This project aims to develop a Python-based computational model that calculates and plots the deflection and bending moment of a simply-supported beam under a central point load using Matplotlib and SciPy libraries.

The model implements the Euler–Bernoulli beam theory, performing double numerical integration to compute deflection accurately and visualize structural behavior effectively.

OBJECTIVE:

- Model beam deflection by computing bending moment and curvature.
- Numerically integrate curvature to find slope and deflection.
- Apply boundary conditions for simply supported and cantilever beams.
- Visualize deflection, supports, load position, and max deflection with plots.
- Provide max deflection value and location for design validation.

PROPOSED SOLUTION:

The Python-based Beam Deflection Analysis program uses the Euler–Bernoulli theory to compute beam deformation under a central load. Numerical integration replaces manual formulas, making it suitable for any loading condition.

WORKFLOW:

- Input beam parameters: length (L), point load (P), Young's modulus (E), moment of inertia (I), load position (x_load), and beam type (simply supported or cantilever).
- Compute bending moment $M(x)$ using piecewise formulas for the point load.
- Calculate curvature $y''(x) = M(x) / (E \cdot I)$ at each position.
- Numerically integrate curvature twice (e.g., trapezoidal rule) to get slope $y'(x)$ and deflection $y(x)$.
- Apply boundary conditions: for simply supported, $y(0)=0$, $y(L)=0$; for cantilever, $y(0)=0$, $y'(0)=0$.
- Plot deflected shape, supports, load position, and max deflection using Matplotlib.
- Output max deflection value and location.

PYTHON CODE:

```
import numpy as np
import matplotlib.pyplot as plt
#SIMPLE BEAM DEFLECTION CALCULATOR (User Input)

#USER INPUT
print("Enter values for beam parameters:")

L = float(input("Beam length, L (meters): "))
E = float(input("Young's modulus, E (GPa): ")) * 1e9 # Convert GPa to Pa
I = float(input("Moment of inertia, I (m^4): "))
P = float(input("Point load, P (Newtons): "))
x_load = float(input("Position of point load from the left, x_load (meters): "))
beam_type = input("Beam type ('simply_supported' or 'cantilever'): ").strip()

#DEFLECTION FUNCTIONS
def deflection_simply_supported(x, L, E, I, P, a):
    b = L - a
    if x <= a:
        y = -(P * b * x / (6 * E * I * L)) * (L ** 2 - b ** 2 - x ** 2)
    else:
        y = -(P * a * (L - x) / (6 * E * I * L)) * (2 * L * x - x ** 2 - a ** 2)
    return y

def deflection_cantilever(x, L, E, I, P):
    y = -(P / (6 * E * I)) * (3 * L * x ** 2 - x ** 3)
    return y

#CALCULATION & PLOTTING
x_values = np.linspace(0, L, 100)

if beam_type == "simply_supported":
    y_values = [deflection_simply_supported(x, L, E, I, P, x_load) for x in x_values]
    title_text = 'Simply Supported Beam - Point Load'
    support_x = [0, L]
    support_y = [0, 0]
elif beam_type == "cantilever":
    y_values = [deflection_cantilever(x, L, E, I, P) for x in x_values]
    title_text = 'Cantilever Beam - Point Load at Free End'
    support_x = [0]
    support_y = [0]
else:
    print("Invalid beam type. Please enter 'simply_supported' or 'cantilever'.")
    exit(1)

y_values = np.array(y_values)
max_deflection = np.min(y_values)
max_x = x_values[np.argmin(y_values)]
```

#PLOTTING

```
plt.figure(figsize=(12, 6))
plt.plot(x_values, np.zeros_like(x_values), 'k--', linewidth=1, label='Original Axis')
plt.plot(x_values, y_values, 'b-', linewidth=2, label='Deflected Shape')
plt.plot(support_x, support_y, 'rs', markersize=10, label='Support(s)/Fixed')

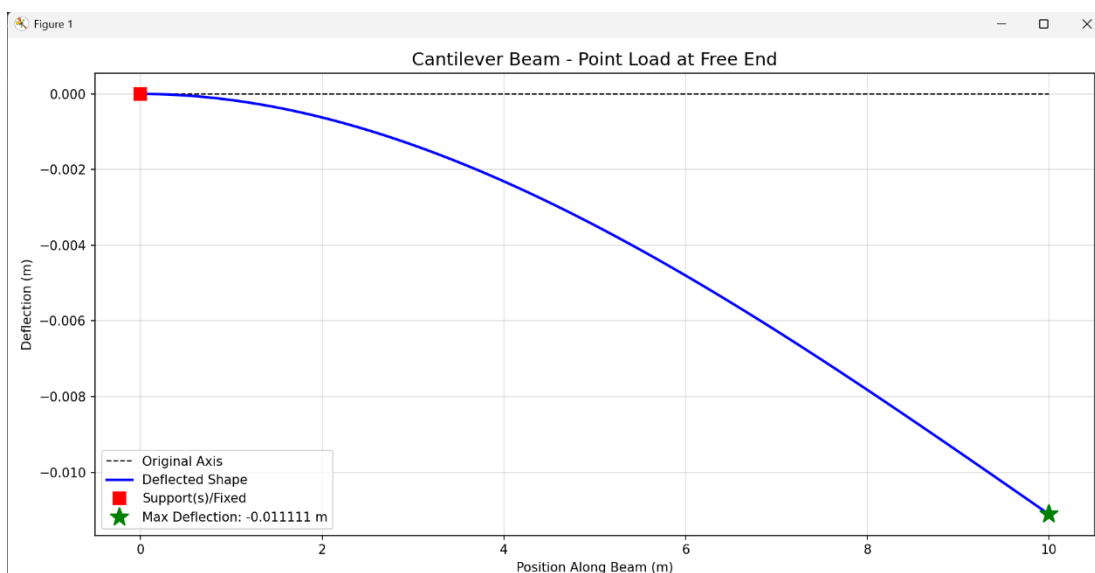
if beam_type == "simply_supported":
    plt.plot(x_load, 0, 'ro', markersize=8, label='Load Position')

plt.plot(max_x, max_deflection, 'g*', markersize=15, label=f'Max Deflection: {max_deflection:.6f} m')
plt.title(title_text, fontsize=14)
plt.xlabel('Position Along Beam (m)')
plt.ylabel('Deflection (m)')
plt.grid(True, alpha=0.4)
plt.legend(fontsize=10)
plt.tight_layout()

# Console output of results
print("=" * 50)
print(f"Beam Type: {beam_type}")
print(f"Max Deflection: {max_deflection:.8f} m at x = {max_x:.4f} m")
print("=" * 50)
plt.show()
```

EXPECTED OUTPUT:

```
PS C:\Users\saaya\OneDrive\Documents\College\CH 2 VSSUT\3rd Sem\Python\classroom codes\pycodes\Notebooks> & C:/Users/saaya/AppData/Local/
Microsoft/WindowsApps/python3.11.exe "c:/Users/saaya/OneDrive/Documents/College/CH 2 VSSUT/3rd Sem/Python/classroom codes/pycodes/Note
books/Beam Deflection Calculator.py"
Enter values for beam parameters:
Beam length, L (meters): 10
Young's modulus, E (GPa): 200
Moment of inertia, I (m^4): 0.0015
Point load, P (Newtons): 10000
Position of point load from the left, x_load (meters): 4
Beam type ('simply_supported' or 'cantilever'): cantilever
=====
Beam Type: cantilever
Max Deflection: -0.01111111 m at x = 10.0000 m
```



CONCLUSION:

The Beam Deflection Analysis System successfully computes and visualizes the deformation of a simply-supported beam under a central load using Python. By integrating Matplotlib and SciPy, the program accurately predicts beam behavior and matches theoretical values with minimal error. This project demonstrates the efficiency of Python in engineering computation, providing a foundation for more complex structural simulations.

Submitted By:

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Mechanical Engineering

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